LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Static, dynamic and total pressure, principle of A.S.I, Indicated, rectified, equivalent and true air speed, errors in A.S.I.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Static, dynamic and total pressure, principle of A.S.I, Indicated, rectified, equivalent and true air speed, errors in A.S.I.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Pitot-Static tube installation; Safety: Ensure no loose parts; Supervision: Instructor monitors probe alignment.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of static pressure?
(b) How does dynamic pressure relate to airspeed?
(c) What is the function of the pitot-static system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of static pressure • Definition of dynamic pressure • Definition of total pressure • Bernoulli's principle application • Pitot-static tube operation	L & Ds	PPT, IB	15 min
(b)	• Indicated airspeed definition • Rectified airspeed correction • Equivalent airspeed calculation • True airspeed derivation • Altitude and temperature effects	L & Ds	PPT, PDF	20 min
(c)	• Position error sources • Instrument error types • Compressibility error impact • Calibration procedures • Error correction methods	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between indicated and true airspeed.
(b) Identify two common sources of ASI error.
(c) Describe how temperature affects equivalent airspeed.

12. Home Assignment and Follow Up	Review the definition of streamline and stream tube from the previous lesson plan to understand flow characteristics relevant to pressure distribution.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of streamline, stream tube, stream filament, ideal and real fluid, steady and turbulent flow, compressible and incompressible flow, viscous and in viscous flow, rotational and irrotational flow.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min